

Machine Learning

Dimensionality Reduction

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我们是否应该讲授深度学习之前的传统方法？还是直接通过深度学习解决所有的问题？



深度学习 vs 传统方法

Deep vs. Classical Methods

Some topics for discussion

Richard Szeliski

The University of Washington



ICCV 21 Panel on Deep vs. Classical Methods

October 13, 2021

A discussion about deep learning vs classical methods and their roles in computer vision

Aude Oliva (MIT-IBM Watson 人工智能实验室)、Svetlana Lazebnik (伊利诺伊大学香槟分校)、Jitendra Malik (加州大学伯克利分校)、Andrew Davison (帝国理工大学)、Richard Szeliski (华盛顿大学)、Alexei Efros (加州大学伯克利分校)。

Contents

- Dataset
- Preprocessing
- Dimensionality Reduction

You need datasets

You can use open datasets

For example for experimenting a new ML algorithm:

- UCI ML Repo : <http://archive.ics.uci.edu/ml/>
- Kaggle competitions, e.g. <https://www.kaggle.com/c/diabetic-retinopathy-detection>
- specific well known datasets for specific ML problems

Some available datasets

Face expression classification

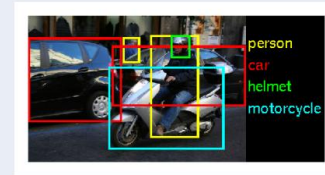
48x48 pixel grayscale images of faces
0=Angry, 1=Disgust, 2=Fear, 3=Happy, 4=Sad, 5=Surprise, 6=Neutral
28K Train; 3K for public test, another 3K for final test.



Kaggle, ICML 2013

Object localization/detection

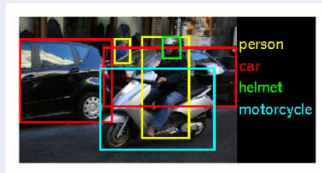
ImageNet, ILSVRC2014: 1000 classes, 1.2M Train images, 50K Valid, 100K Test
Avg image size : 482x415 pixels, RGB



ImageNet Large Scale Visual Recognition Challenge, Russakovsky et al. (2015)

Object localization/detection

ImageNet, ILSVRC2014: 1000 classes, 1.2M Train images, 50K Valid, 100K Test
Avg image size : 482x415 pixels, RGB



ImageNet Large Scale Visual Recognition Challenge, Russakovsky et al. (2015)

Object segmentation

COCO 2017: 200K images, 80 classes, 500K masks



<http://cocodataset.org/>

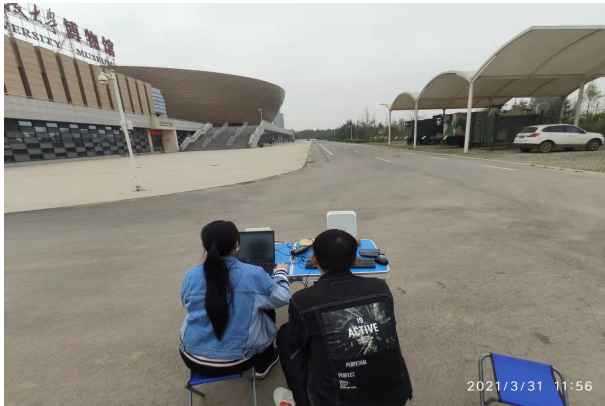
You need datasets ! !

You have a specific problem

You may need to collect data on your own.

- Crawl the web ? (e.g. Tweeter API, ..)
- if supervised learning : assign labels (mechanical turk, domain experts (classifying tumors))
- Ensure you collected sufficient features

You need datasets ! !



Preprocessing data

Data are not necessarily vectorial

- Ordinal or Categorical : poor/faire/excellent ; Male/Female
- Text documents : bag of words / word embeddings

Even if vectorial

- Missing data : check how missing values are indicated (-9, ' ', ..) → Imputation of missing values
- Feature scaling

Preprocessing data

- Your data might not be vectorial data

Ordinal and categorical features

Ordinal values have an order.

Ordinal Feature value	poor	fair	excellent
Numerical feature value	-1	0	1

Categorical values do not have an order (use one-hot) :

Categorical value	American	Spanish	German	French
Numerical value	[1, 0, 0, 0]	[0, 1, 0, 0]	[0, 0, 1, 0]	[0, 0, 0, 1]

Preprocessing data

Vectorial representation of text documents

Bag Of Words

- define a vocabulary $\mathcal{V}$, $|\mathcal{V}| = n$
- for each document, build a vector x so that x_i is the frequency of the word $\mathcal{V}_i$

e.g. $\mathcal{V} = \{I, in, love, metz, machinelearning, study\}$

I love machine learning and love metz too. $\rightarrow x = [1, 0, 2, 1, 1, 0]$

I love studying machine learning in Metz. $\rightarrow x = [1, 1, 1, 1, 1, 1]$

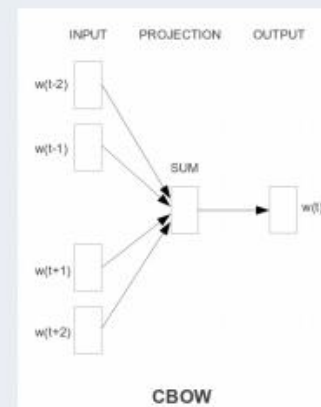
Does not take the order into account $\rightarrow N - gram$, but this leads to sparser representations

Preprocessing data

Vectorial representation of text documents

Word/Sentence embeddings (e.g. word2vec, GLoVe, fasttext).
Continuous Bag of Words (CBOW) : predict a word given its context

- Input and output coded with one-hot
- predict a word given its context
- hidden layer : word representation



Captures some semantic information.

For sentences : tweet2vec, sentence2vec, word vector avg

Preprocessing data

- Some features might be missing

Missing features

- Completely **drop out** the samples with missing attributes, or the dimensions that have missing values
- or try to **impute**, i.e. set a value in place of the missing attributes

For missing value imputation, there are plenty of methods :

- global : assign the mean, median, most frequent value of an attribute
- local : based on k-nearest neighbors, decide which value to impute

The bias you may introduce by imputing a value may depend on the causes of the missing values, see [Silva(2014)].

Silva(2014). A brief review of the main approaches for treatment of missing data

Preprocessing data

- Some vectorial data might not be appropriately scaled

Feature scaling

- dimensions with the largest variations will dominate euclidean distances (e.g. nearest neighbors)
- when gradient descent is involved, feature scaling makes convergence faster (because the loss is circular symmetric)
- when regularization is involved, we would like to use a single regularization coefficient, independent on the scale of the features

Preprocessing data

Feature scaling

Given $x_i \in \mathbb{R}^d$, you can normalize by :

- min/max scaling :

$$\forall i, \forall j \in [0, d-1] x'_{i,j} = \frac{x_{i,j} - \min_k x_{k,j}}{\max_k x_{k,j} - \min_k x_{k,j}}$$

- z-score normalization :

$$\forall i, \forall j \in [0, d-1] x_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j}$$

$$\mu_j = \frac{1}{N} \sum_k x_{k,j}$$

$$\sigma_j = \sqrt{\frac{1}{N} \sum_k (x_{k,j} - \mu_j)^2}$$

Your statistics must be computed from the training set and applied also to test data

Dimensionality reduction : what/why/how

What

Optimally transform $x_i \in \mathbb{R}^n$ into $z_i \in \mathbb{R}^d$ so that $d \ll n$
It remains to define what means “optimally transform”

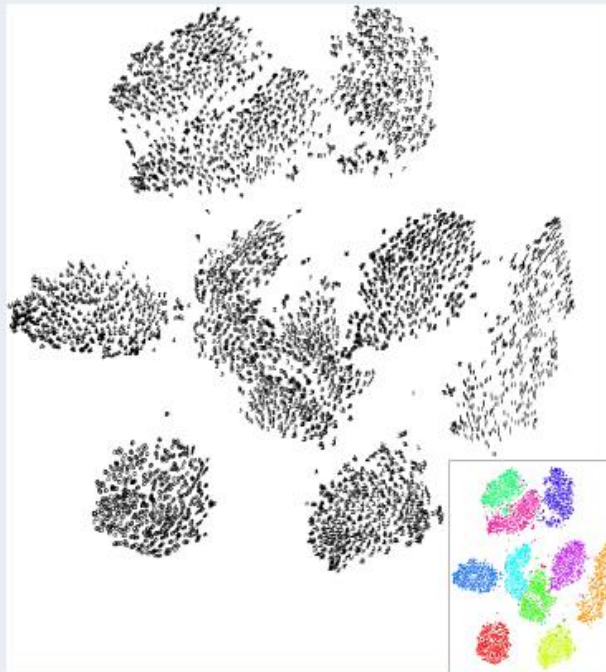
Why

- visualization of the data
- interpretability of the predictor
- speed up the algorithms whose complexity depends on n
- data may occupy a manifold of lower dimensionality than n
- curse of dimensionality : data get quickly sparse, models may overfit

Dimensionality reduction: why ?

Data analysis/Visualization

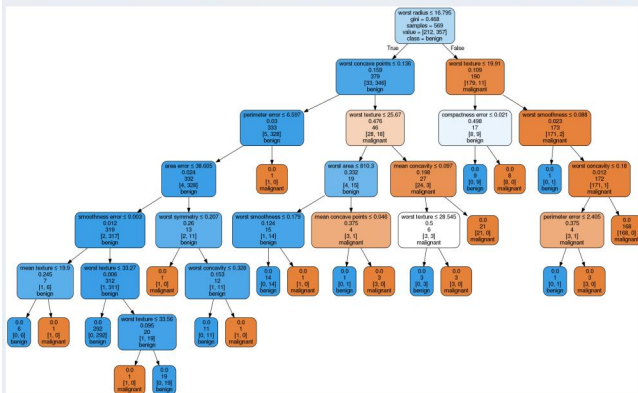
How are your data distributed ? How are your classes intricate ?
Do we have discriminative features ?



Dimensionality reduction: why ?

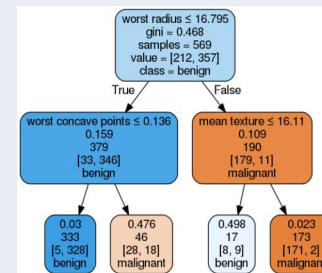
Interpretability of the predictor

e.g. Why does this predictor say the tumor is malignant ?



Real risk = 0.92 ± 0.05

UCI ML Breast Cancer Wisconsin (Diagnostic) dataset
Real risk estimated by 10-fold CV.



Real risk = 0.92 ± 0.06

Dimensionality reduction: why ?

Speed up of the algorithms

Decreasing dimensionality decreases training/inference times.

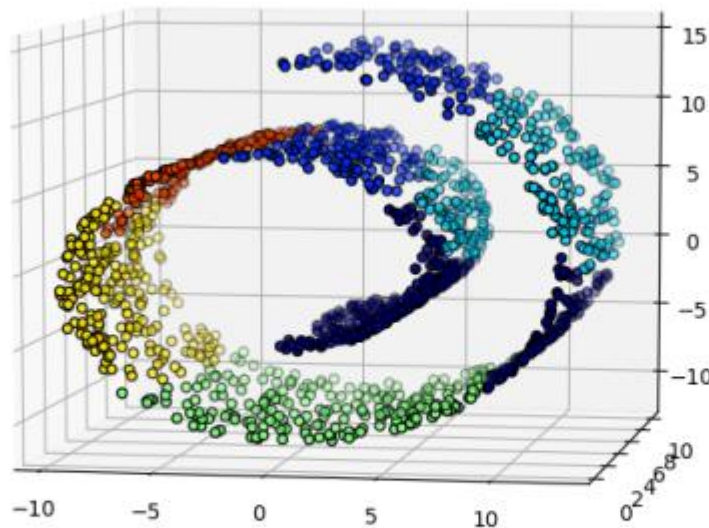
For example :

- Linear regression $\hat{y} = \theta^T x + b$
- Logistic regression(classification) : $P(y = 1/x) = \frac{1}{1+\exp(\theta^T x)}$

Both training and inference in $O(n)$, $x \in \mathbb{R}^n$

Dimensionality reduction: why ?

The data may occupy a lower dimensional manifold



Swiss roll

→ you do not necessarily lose information by reducing the number of dimensions

Dimensionality reduction: why ?

The data may occupy a lower dimensional manifold

You want to classify **facial expressions** of a **single person**, controlled illumination:

- suppose a huge image resolution, e.g. 1024×1024 RGB pixels, $x \in \mathbb{R}^{1024 \times 1024 \times 3}$
- what is the dimensionality of the data manifold ? ≈ 50



Dimensionality reduction: why ?

The data may occupy a lower dimensional manifold

You want to classify **facial expressions** of a **single person**, controlled illumination:

- suppose a huge image resolution, e.g. 1024×1024 RGB pixels, $x \in \mathbb{R}^{1024 \times 1024 \times 3}$
- what is the dimensionality of the data manifold ? ≈ 50

→ you do not necessarily lose information by reducing the number of dimensions

Dimensionality reduction: why ?

The data may occupy a lower dimensional manifold

You want to classify **facial expressions** of a **single person**, controlled illumination:

- suppose a huge image resolution, e.g. 1024×1024 RGB pixels, $x \in \mathbb{R}^{1024 \times 1024 \times 3}$
- what is the dimensionality of the data manifold ? ≈ 50

→ you do not necessarily lose information by reducing the number of dimensions

Dimensionality reduction: why ?

You may even have better predictors : Curse of dimensionality

The data become (exponentially) quickly sparse with respect to the dimension

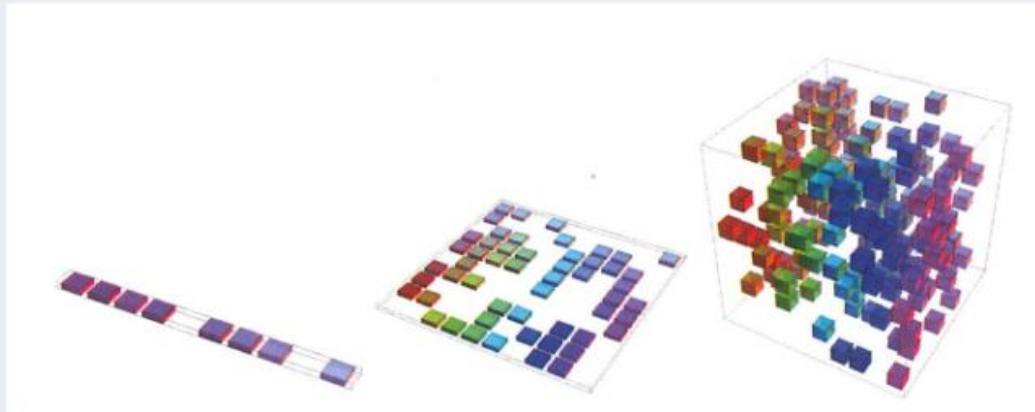


Image from [Goodfellow, Bengio, Courville (2016) : Deep learning]
See also [Hastie et al.(2017), The elements of statistical learning]

Dimensionality reduction : what/why/how ?

What

Optimally transform $x_i \in \mathbb{R}^n$ into $z_i \in \mathbb{R}^d$ so that $d \ll n$
It remains to define what means “optimally transform”

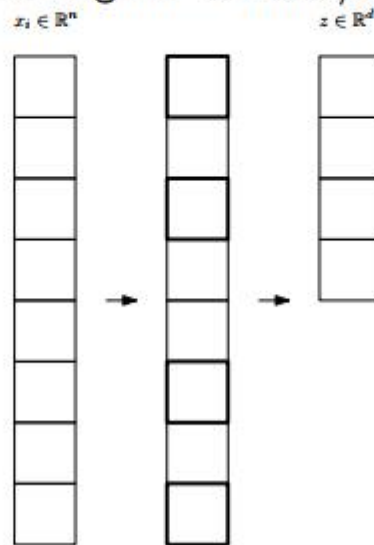
How

- select a subset of the original features : **feature selection**
- compute new features from the original ones : **feature extraction**

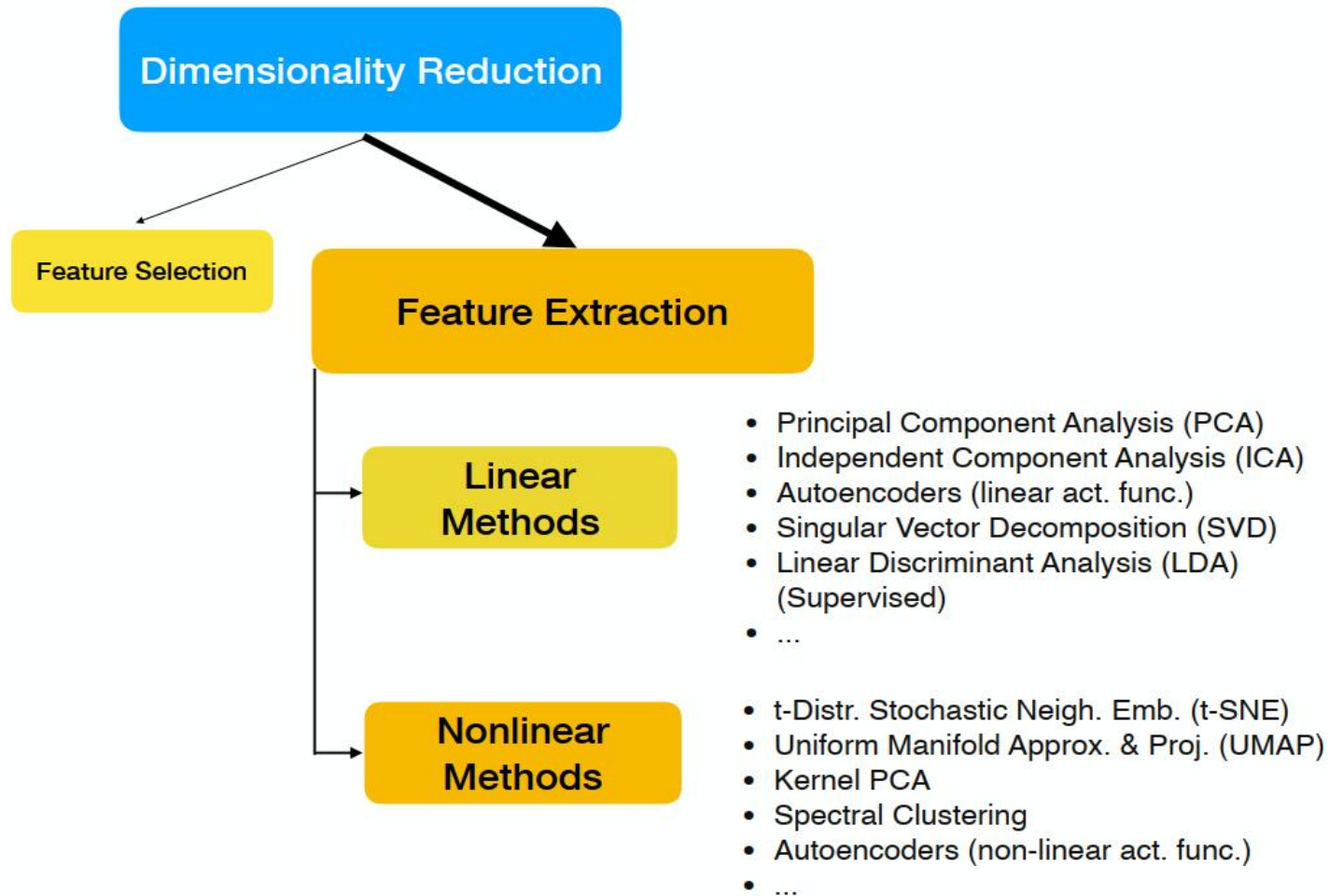
Dimensionality reduction : what/why/how ?

Feature selection

Select a subset of the original features/attributes/dimensions



Learning?



Feature extraction

Given N samples $x_i \in \mathbb{R}^d$,

We compute $r \ll d$ new features from the original d features.



A word cloud of feature extraction techniques. The words are arranged in a circular pattern, with 'Sammon's' at the top left, 'mapping' at the top right, 'Visualization' in the center, 'Manifold' below it, 't-SNE' on the left, 'Autoencoders' on the right, 'Isomap' at the bottom left, 'LDA' in the center, 'PCA' on the right, 'LLE' below 'PCA', and 'kernel' at the bottom. The words are in various colors including red, orange, blue, and black.

Sammon's
mapping
Visualization
Manifold
t-SNE
Autoencoders
Isomap
LDA
PCA
LLE
kernel

Principal Component Analysis (PCA)

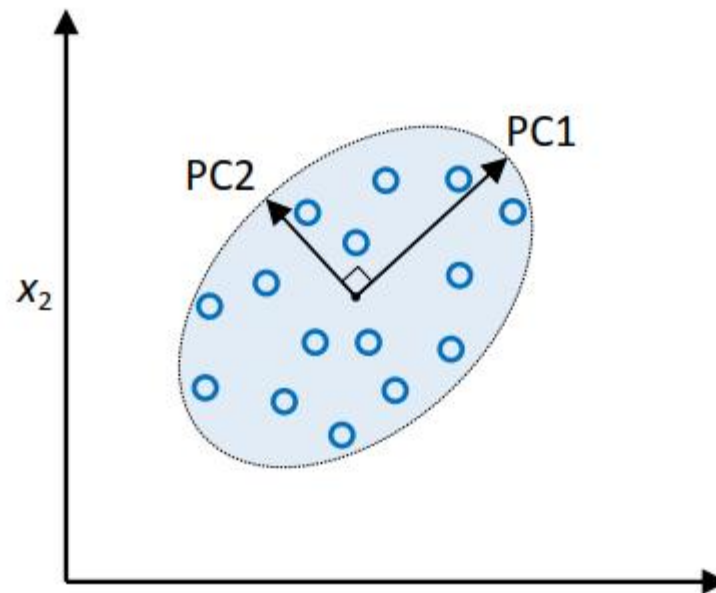
Equivalent formulations

There are two equivalent formulations of the PCA :

- Find an affine transformation minimizing the reconstruction error
- Find an affine transformation maximizing the variance of the projections

Principal Component Analysis (PCA)

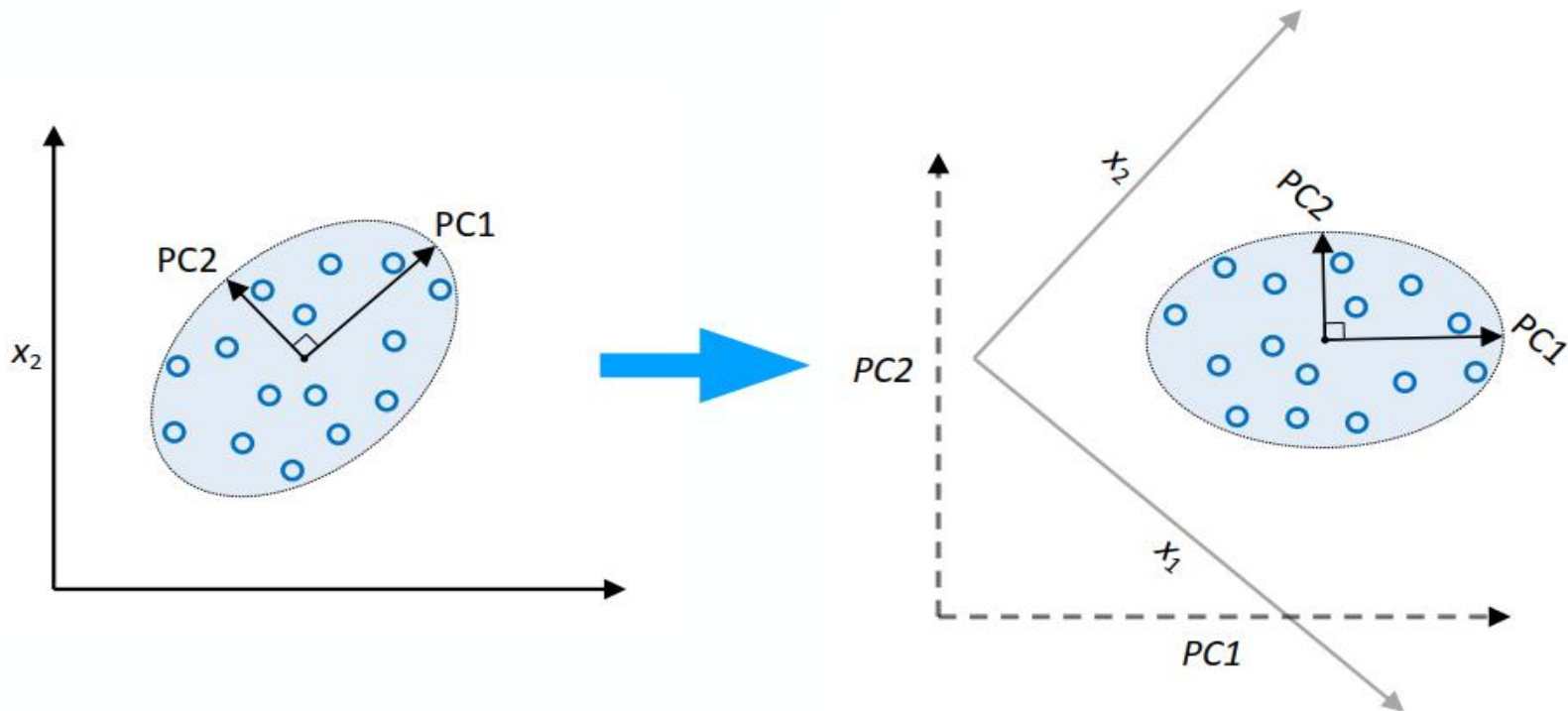
1) Find directions of maximum variance



Find an affine transformation of the data minimizing the reconstruction error

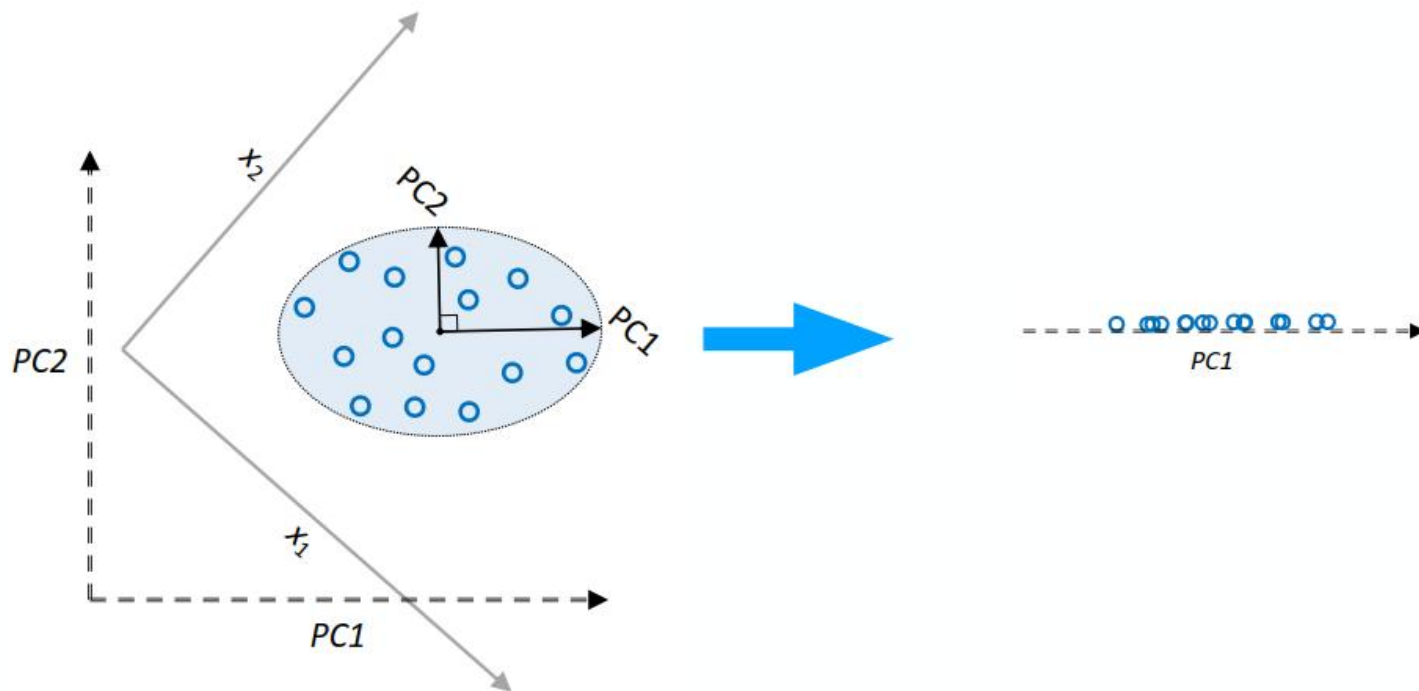
Principal Component Analysis (PCA)

2) Transform features onto directions of maximum variance



Principal Component Analysis (PCA)

3) Usually consider a subset of vectors of most variance (dimensionality reduction)



Principal Component Analysis (PCA)

Given design matrix $\mathbf{X} \in \mathbb{R}^{n \times m}$

find vector α_i with maximum variance

repeat: find α_{i+1} with maximum variance uncorrelated with α_i

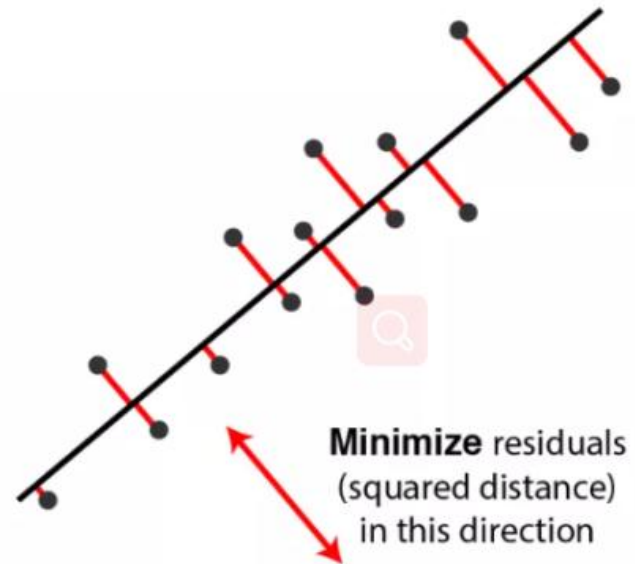
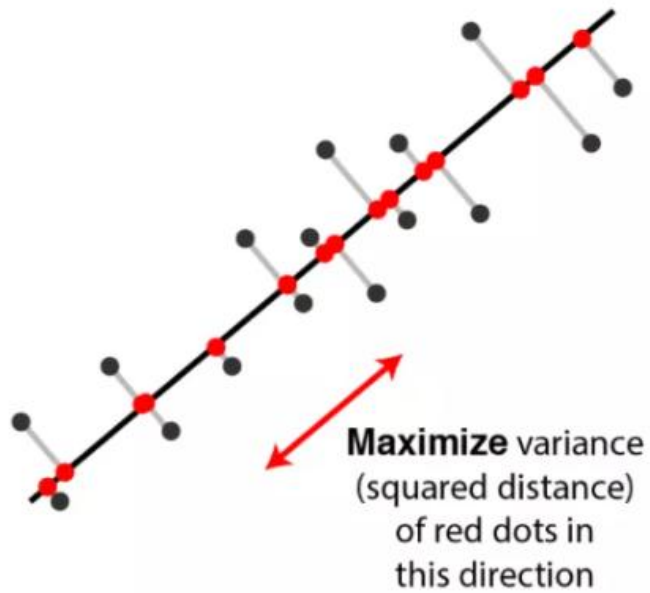
(repeat k times, where k is the desired number of dimensions; $k \leq m$)

Principal Component Analysis (PCA)

- Two approaches to solve PCA (on standardized data):
 - 1. Constrained maximization (e.g., Lagrange multipliers)
 2. Eigen-decomposition of covariance matrix directly

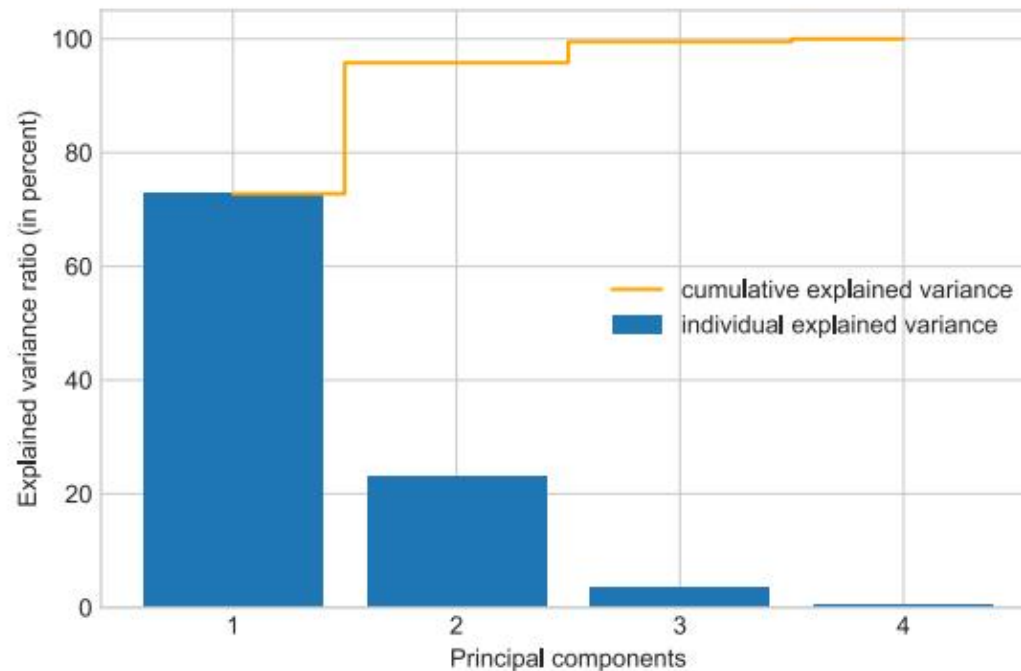
Collect vectors α_i in a projection matrix $\mathbf{A} \in \mathbb{R}^{m \times k}$
(Sorted from highest to lowest associated eigenvalue)

Compute projected data points: $\mathbf{Z} = \mathbf{XA}$

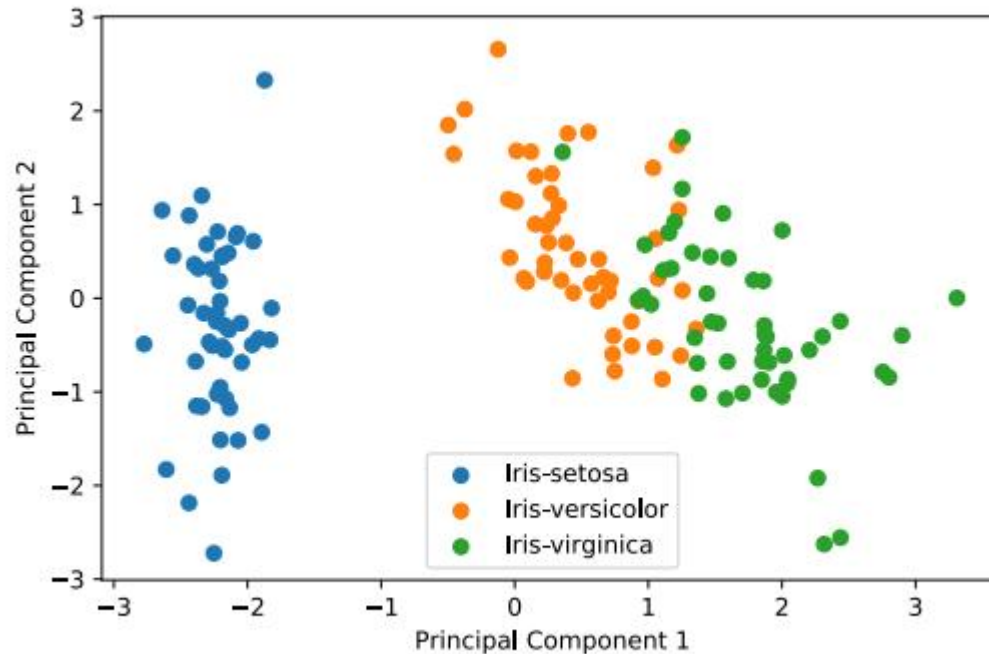


Principal Component Analysis (PCA)

Usually useful to plot the explained variance (normalized eigenvalues)



Principal Component Analysis (PCA)



Keep in mind that PCA is *unsupervised*!

Feature extraction : Manifold learning

Goal : For each $x_i \in \mathbb{R}^d$, associate $y_i \in \mathbb{R}^r$ so that the pairwise distances in $\mathbb{R}^d$ are **as similar as possible** to the pairwise distance in $\mathbb{R}^r$

Perfect for visualizing the datasets in low dimensions.

Examples : LLE, MDS, Isomap, SNE, t-SNE, ..

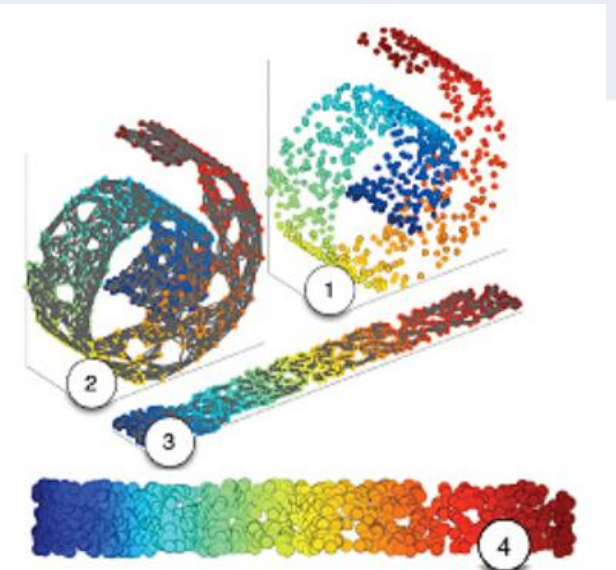
Nearby point remain nearby
Distant points remain distant

Manifold learning

Overview

$x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}^r$, $r \ll d$, e.g. $r = 2$

- ❶ Quantify the similarity between pairs of points in $\mathbb{R}^d$
- ❷ Quantify the similarity between pairs of points in $\mathbb{R}^r$
- ❸ Quantify the discrepancy between these similarities
- ❹ Optimize with respect to y_i



Manifold learning

Rank ordering of Euclidean distances is
NOT preserved in “manifold learning”.



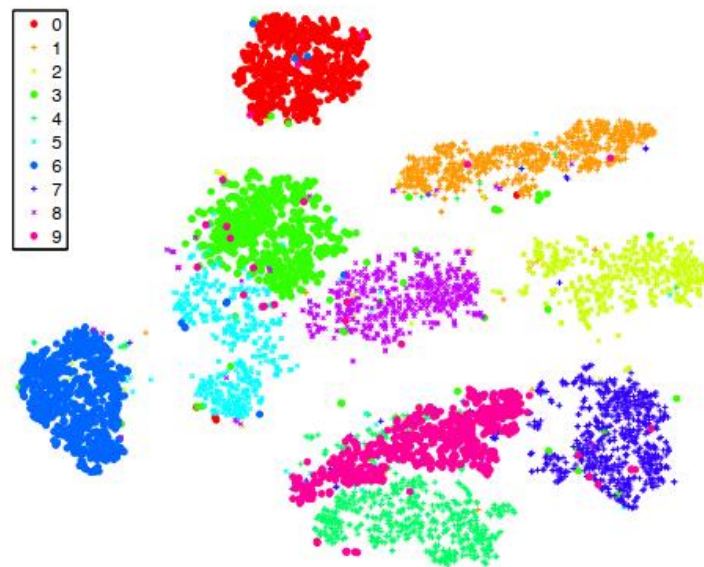
$$d(A,C) < d(A,B)$$



$$d(A,C) > d(A,B)$$

t-Distributed Stochastic Neighbor Embedding (*t-SNE*)

(t-SNE is only meant for visualization not for preparing datasets!)



Note that MNIST has
 $28 \times 28 = 784$ dimensions

(a) Visualization by t-SNE.

Shown are 6000 images from MNIST projected in 2D

Maaten, L. V. D., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of machine learning research*, 9(Nov), 2579-2605.



t-Distributed Stochastic Neighbor Embedding (*t-SNE*)

Given high-dimensional datapoints, $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^m$
represent intrinsic structure of the data in 1D, 2D, or 3D (for visualization)

How?

- 1) Model neighboring datapoint pairs based on the distance of those points in the high-dimensional space
- 2) Find a probability distribution of the pairwise distances in the low dimensional space that is as close as possible as the original probability distribution

Main Idea: Map points near on a manifold to a near position in low-dimensional space

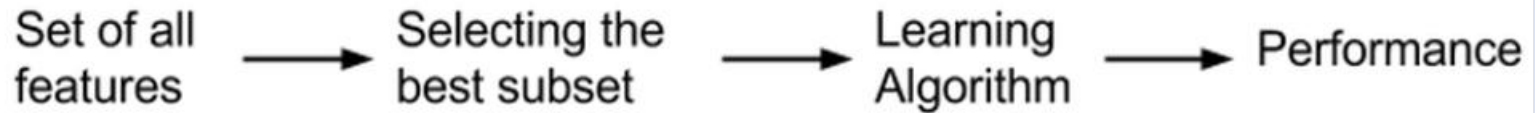
Feature selection

What's/why feature selection

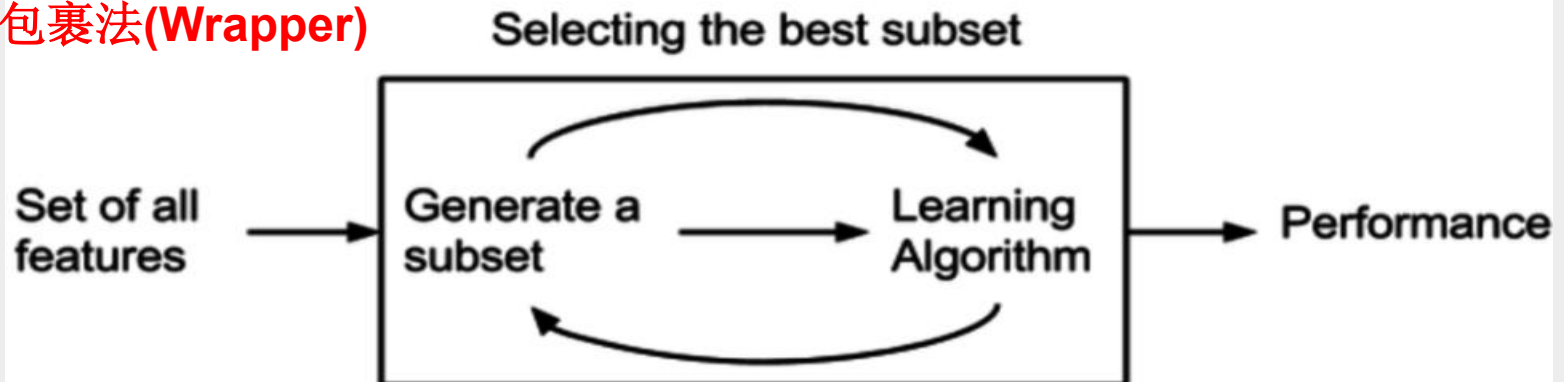
- A procedure in machine learning to find a subset of features that produces 'better' model for given dataset
 - Avoid overfitting and achieve better generalization ability
 - Reduce the storage requirement and training time
 - Interpretability

When feature selection is important

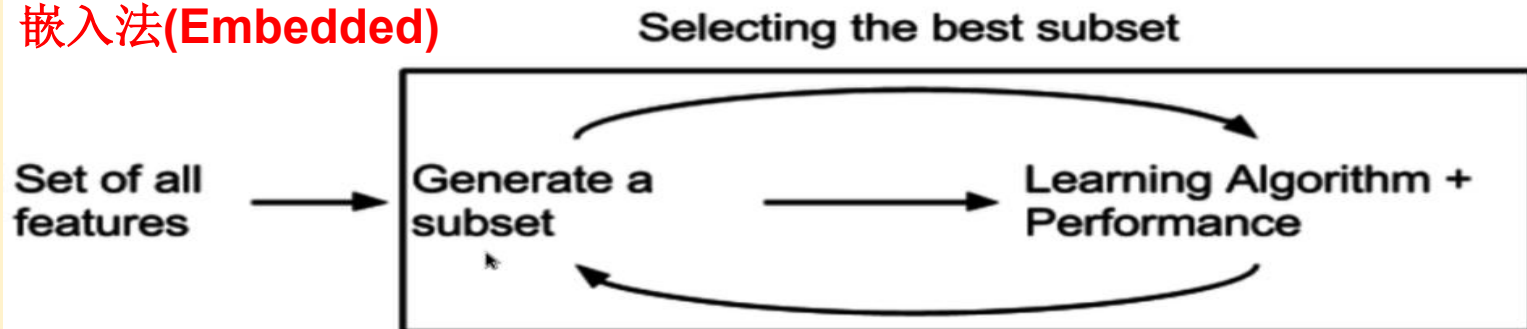
- Noise data
- Lots of low frequent features
- Use multi-type features
- Too many features comparing to samples
- Complex model
- Samples in real scenario is inhomogeneous with training & test samples



包裹法(Wrapper)

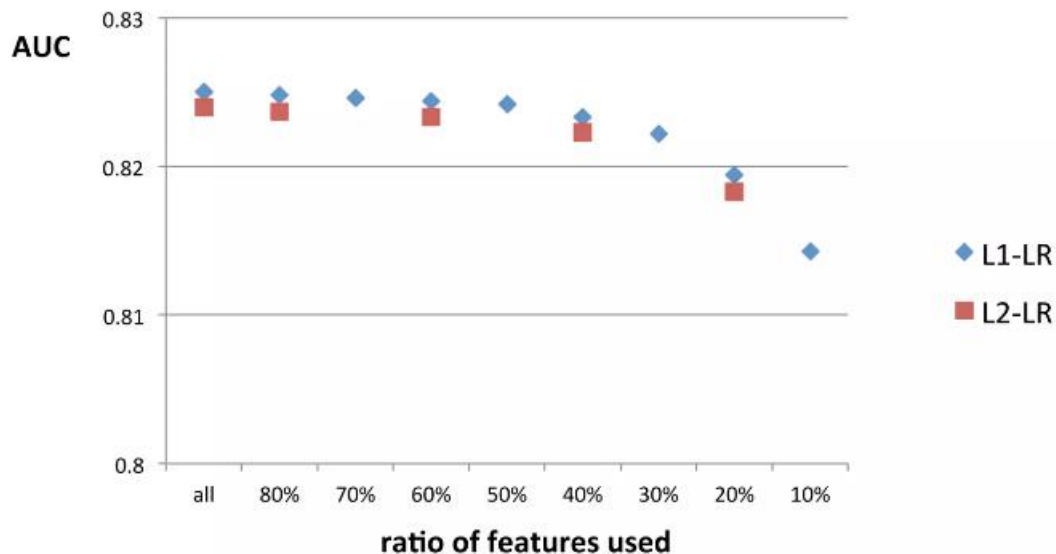


嵌入法(Embedded)



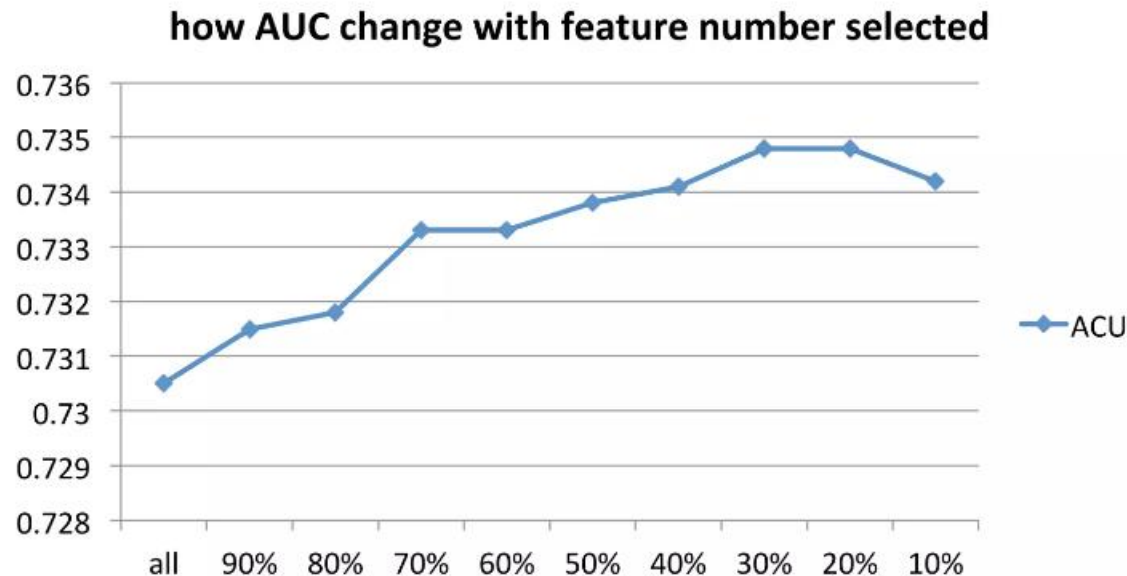
When No.(samples)/No.(feature) is large

- Feature selection with **Gini indexing**
- Algorithm: Logistic regression
- Training samples: 640k; test samples: 49K
- Feature: watch behavior of audiences; **show level** (11327)



When No(samples) equals to No(feature)

- L1 Logistic regression;
- Training samples: 50k; test samples: 49K
- Feature: watch behavior of audiences; **video level** (49166)



Typical methods for feature selection

- Categories

	Single feature evaluation	Subset selection
filter	MI, IG, KL-D, GI, CHI	Category distance, ...
wrapper	Ranking accuracy using single feature	For LR (SFO, Grafting)

- Single feature evaluation

- Frequency based, mutual information, KL divergence, Gini indexing, information gain, Chi square statistic

- Subset selection method

- Sequential forward selection
- Sequential backward selection

Single feature evaluation

- Measure quality of features by all kinds of metrics
 - Frequency based
 - Dependence of feature and label (Co-occurrence)
 - mutual information, Chi square statistic
 - Information theory
 - KL divergence, information gain
 - Gini indexing

Frequency based

- Remove features according to frequency of features or instances contain the feature
- Typical scenario
 - Text mining

Mutual information

- Measure the dependence of two random variables

- Definition

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \left(\frac{p(x, y)}{p(x) p(y)} \right)$$

$$I(X; Y) = \int_Y \int_X p(x, y) \log \left(\frac{p(x, y)}{p(x) p(y)} \right) dx dy$$

Chi Square Statistic

- Measure the dependence of two variables

$$\chi^2(t, c) = \frac{N \times (AD - CB)^2}{(A + C) \times (B + D) \times (A + B) \times (C + D)}$$

- A: number of times feature t and category c co-occur
- B: number of times t occurs without c
- C: number of times c occurs without t
- D: number of times neither c or t occurs
- N: total number of instances

Entropy

- Characterize the (im)purity of an collection of examples

$$Entropy(S) = -\sum_i P_i \ln P_i$$



Information Gain

- Reduction in entropy caused by partitioning the examples according to the attribute

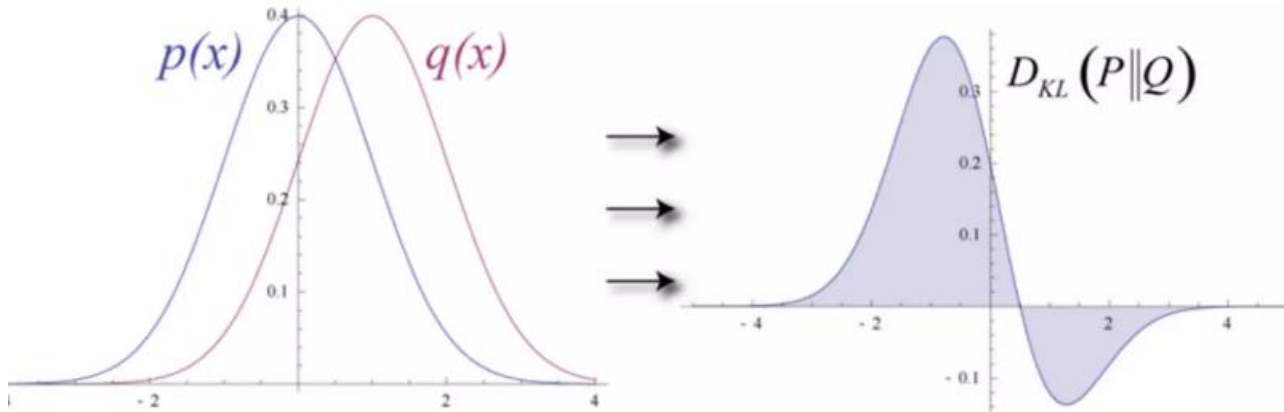
$$Gain(S, f) = Entropy(S) - (\Pr(f)Entropy(S | f) + \overline{\Pr(f)}Entropy(S | \bar{f}))$$

$$IG(t) = -\sum_i Pr(c_i) \log Pr(c_i) + Pr(t) \sum_i Pr(c_i|t) \log Pr(c_i|t) \\ + Pr(\bar{t}) \sum_i Pr(c_i|\bar{t}) \log Pr(c_i|\bar{t})$$

KL divergence

- Measure the difference between two probability distribution

$$D_{\text{KL}}(P\|Q) = \sum_i P(i) \ln \frac{P(i)}{Q(i)}$$



$$\begin{aligned} D_{\text{KL}}(P\|Q) &= -\sum_x p(x) \log q(x) + \sum_x p(x) \log p(x) \\ &= H(P, Q) - H(P) \end{aligned}$$

Gini indexing

- Calculate conditional probability of f given class label

$$Pr(f|c_i) = \frac{1 + \mathcal{N}(f, c_i)}{|V| + \sum_{f \in V} \mathcal{N}(f, c_i)}$$

- Normalize across all classes

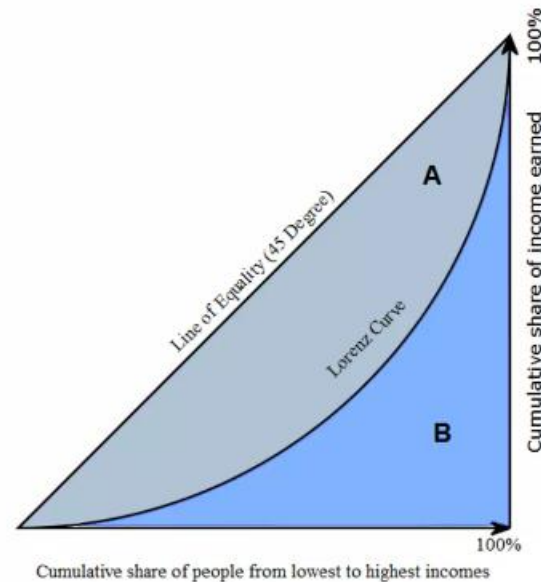
$$wcp(f, c_i) = \frac{Pr(f|c_i)}{\sum_{k=1}^{|C|} Pr(f|c_k)}$$

- Calculate Gini coefficient

$$\mathcal{G}(t) = \frac{\sum_{i=1}^n (2i - n - 1) wcp(t, c_i)}{n^2 \mu}$$

- For two categories case

$$\varsigma(t) = \frac{wcp(t, c_2) - wcp(t, c_1)}{2(wcp(t, c_2) + wcp(t, c_1))}$$

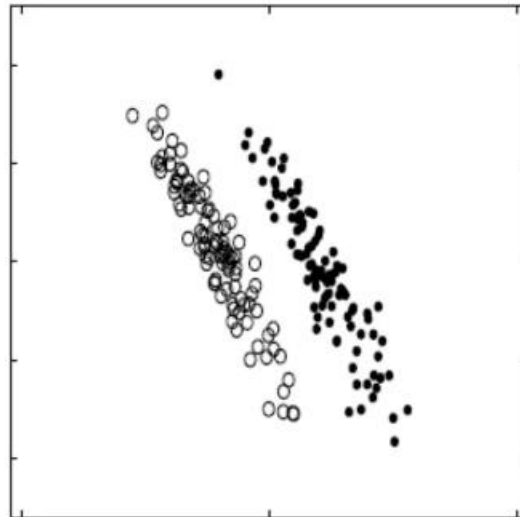


Shortages of single feature evaluation

- Relevance between features are ignored
 - Features could be redundant
 - A feature that is completely useless by itself can provide a significant performance improvement when taken with others
 - Two features that are useless by themselves can be useful together

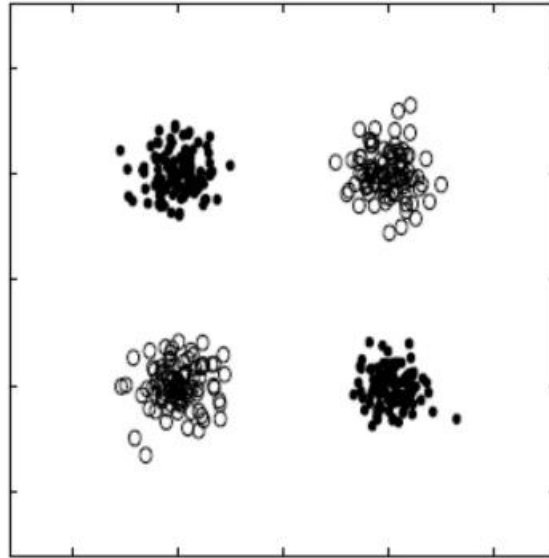
Shortages of single feature evaluation

- A feature that is completely useless by itself can provide a significant performance improvement when taken with others



Shortages of single feature evaluation

- Two features that are useless by themselves can be useful together



Subset selection methods

- Select subsets of features that together have good predictive power, as opposed to ranking features individually
- Always by adding new features into existing set or removing features out of existing set
 - Sequential forward selection
 - Sequential backward selection
- Evaluation
 - category distance measurement
 - Classification error

- Select features subset with large category distance

$$J(x) = \frac{1}{2} \sum_{i=1}^C P_i \sum_{j=1}^C P_j \frac{1}{n_i n_j} \sum_{k=1}^{n_i} \sum_{l=1}^{n_j} \ell(x_k^{(i)}, x_l^{(j)})$$

$$\ell(x_k^{(i)}, x_l^{(j)}) = (x_k^{(i)} - x_l^{(j)})^T (x_k^{(i)} - x_l^{(j)})$$

Wrapper methods for logistic regression

- Forward feature selection
 - Naïve method
 - need build models quadratic in the number of feature
 - Grafting
 - Single feature optimization

- Categories

	Single feature evaluation	Subset selection
filter	MI, IG, KL-D, GI, CHI	Category distance, ...
wrapper	Ranking accuracy using single feature	For LR (SFO, Grafting)

- Filter + Single feature evaluation
 - Less time consuming, usually works well
- Wrapper + Subset selection
 - Higher accuracy, but easy overfitting

- Remove features could not occur in real scenario
- If no contribution, the less feature the better
- Use L1 regularization for logistic regression
- Use random subspace method

Embedded feature selection

- ▶ In filters and wrappers, feature selection are separated from training;
- ▶ Different to them, embedded feature selection combines both procedures of feature selection and learner training as a whole, and accomplished in the same optimization.
- ▶ In the training procedure of the learner, feature selection has been conducted automatically.

Embedded feature selection

- Recall the regularized linear regression

$$J(\mathbf{w}) = \sum_{i=1}^m \left(y^{(i)} - y(\mathbf{x}^{(i)}, \mathbf{w}) \right)^2 + \lambda \|\mathbf{w}\|_2^2$$

- Replacing the ℓ_2 -norm with the ℓ_1 -norm, we have LASSO (Least Absolute Shrinkage and Selection Operator):

$$J(\mathbf{w}) = \sum_{i=1}^m \left(y^{(i)} - y(\mathbf{x}^{(i)}, \mathbf{w}) \right)^2 + \lambda \|\mathbf{w}\|_1$$